

**WINDEXCO SYRUP**  
(Antitussive - Decongestant - Antihistamine)

**Active Ingredients:**

Each 5ml contains -  
Dextromethorphan HBr ..... 10mg  
Pseudoephedrine HCl ..... 30mg  
Chlorpheniramine Maleate ..... 2mg  
Preservatives  
Methyl Hydroxybenzoate ..... 0.12%w/v  
Propyl Hydroxybenzoate ..... 0.012%w/v

**Presentation:**

Bottles of 90ml and 120ml.

**Product Descriptions:**

Dark brown coloured, cough drop flavoured syrupy solution. Alcohol free.

**Pharmacodynamics:**

**Antitussive:**

Dextromethorphan HBr is a centrally acting cough suppressant. It suppresses the cough reflex by depressing the medullary cough center or associated higher centers. Although it is a congener of narcotic analgesic levorphanol, it has no analgesic or sedative properties. It does depress respiration and is nonaddicting. No evidence of tolerance has been found during long term use.

**Decongestant:**

Pseudoephedrine HCl is a sympathomimetic used for the relief of nasal and bronchial congestion. It acts directly on  $\alpha$ -adrenergic receptors in the respiratory tract mucosa producing vasoconstriction resulting in the shrinkage of swollen nasal mucous membrane, reduction of tissue hyperaemia, oedema, nasal congestion, and an increase in nasal airway potency. Drainage of sinus secretions is increased and obstructed Eustachian ostia may be opened. Relaxation of bronchial smooth muscle by stimulation of  $\beta$ -adrenergic receptor may also occur.

**Antihistamine:**

Chlorpheniramine Maleate is a member of the alkylamine group of antihistamine ( $H_1$  receptor antagonist). It does not block the release of histamine but acts by competitively antagonising the effects of histamine at  $H_1$  receptor sites. If the histamine concentration at the receptor sites exceeds the drug concentration, histamine effects predominate. Histamine released causes allergic reactions; therefore Chlorpheniramine Maleate is used therapeutically for palliative treatment in allergic reactions.

**Pharmacokinetics:**

Dextromethorphan HBr is well absorbed from the gastrointestinal tract after oral administration. It is metabolized by N- and O-demethylation followed by sulphate or glucuronic acid conjugation. The major metabolite is dextrophan, (+)-3-hydroxy-N-methylmorphinan; (+)-3-hydroxymorphinan has also been identified. About 50% of the dose is excreted in the urine in 24 hours, mostly as the glucuronide and sulphate conjugates of the metabolites, about 8% of a dose is excreted as unchanged drug in 6 hours. The estimated minimum lethal dose is 0.5g.

Pseudoephedrine HCl is rapidly and completely absorbed after oral administration. Up to about 90% of a dose is excreted unchanged in the urine in about 24 hours with less than 1% as nor-pseudoephedrine. Plasma half life is between 5 to 8 hours but may be increased when the urine is alkaline or decreased when the urine is acidic. Toxic effects have occurred after single dose of 60mg.

Chlorpheniramine Maleate is readily and almost completely absorbed after oral administration but extensively metabolized to polar and non-polar metabolites. Bioavailability about 35%. The major metabolites are monodesmethyl and didesmethyl-chlorpheniramine. About 35% of a single dose is excreted in the urine in 48 hours; the 24 hours excretion of unchanged drug accounts for about 3 to 10% of the dose but this is increased by acidification of the urine and increased urinary flow and decreased when the urine is alkaline. Toxic effects may be produced by plasma concentration greater than 20 $\mu$ g/ml.

**Indications:**

Symptomatic relief of cough and nasal congestion associated with common cold, allergy and other respiratory infections.

**Dosage and Administration:** For oral administration only.

Adults and children above 12 years: 2 teaspoons (10ml) every 6 hours.

Do not exceed 8 teaspoons or 40ml in a day.

Children 6 to 12 years: 1 teaspoon (5ml) every 6 hours.

Do not exceed 4 teaspoons or 20ml in a day.

Children 2 to 5 years: ½ teaspoon (2.5ml) every 6 hours.

Do not exceed 2 teaspoons or 10ml in a day.

**Symptoms and Treatments of Overdose:**

**Dextromethorphan**

Symptoms: Confusion, unusual excitement, nervousness, restlessness, irritability, respiratory depression. Accidental poisoning (in children) is characterized by stupor and ataxia.

Treatment: The respiratory depression induced by Dextromethorphan may be treated by the use of Naloxone as a specific antagonist. Children will be recovered rapidly from

stupor and ataxia after emesis.

**Chlorpheniramine and Pseudoephedrine**

Symptoms: Weakness, incoordination, difficulty in micturition, respiratory depression, hypotension, convulsion, hypertension, palpitation and tachycardia.

Treatment: Treatment include maintaining and supporting respiration, gastric lavage if indicated and administration of Diazepam to control convulsions. Catheterisation of the bladder may be necessary. Hypertension may be controlled with  $\alpha$ -adrenoceptor blocking drugs and tachycardia with  $\beta$ -adrenoceptor blocking drugs.

**Contraindications:**

Hypersensitivity to Dextromethorphan, Chlorpheniramine or Pseudoephedrine. Marked hypertension and coronary disease.

Not to be used concomitantly nor within 2 weeks of Monoamine Oxidase Inhibitor (MAOI) therapy.

Anticholinergic side effects, eg. dry mouth, blurred vision and constipation.

**Precautions:**

If symptoms persist, please consult a doctor.

*This medicine may cause drowsiness. If affected, do not drive or operate machinery. Avoid alcoholic drinks.*

Cautions in patients with peripheral vascular disease and other cardiovascular disease and patients on antidepressant therapy.

Dextromethorphan should not be given to patient with or at risk of developing respiratory failure.

Administer with caution to patients with liver disease, asthma, diabetes, narrow angle glaucoma, prostatic hypertrophy, cardiac disease and pregnancy.

Caution in patients taking other sympathomimetic agents such as appetite suppressant, decongestant, amphetamine-like psychostimulants or antihypertensive agents.

Not to be given to patients with or at risk of developing respiratory failure.

Not to be used in children less than 2 years of age.

To be used with caution and doctor's / pharmacist's advice in children 2 to 6 years of age.

**Use in Pregnancy:**

Although Pseudoephedrine and Chlorpheniramine have been widely used for many years without apparent ill consequences, it is advisable not to give any drug during the first trimester of pregnancy, unless circumstances demand.

**Use in Lactation:**

Pseudoephedrine and Chlorpheniramine is excreted in breast milk; use by nursing mothers is not recommended because of the higher than usual side-effects from sympathomimetic amines for infants, especially newborn and premature infants.

**Interactions with Other Medicaments:**

Concurrent administration with Fluoxetine or Paroxetine can increase the serum level of both drugs.

Concurrent administration with Quinidine can increase concentration of Dextromethorphan.

Concomitant use of Pseudoephedrine with sympathomimetic agents, such as decongestants, appetite suppressants and amphetamine-like psychostimulants or with MAOIs which interfere with the catabolism of sympathomimetic amines, may occasionally cause a rise in blood pressure, largely because of interaction with Pseudoephedrine.

The antibacterial agent, Furazolidone, is known to cause a dose related inhibition of monoamine oxidase. Although there are no reports of hypertensive crises caused by the concurrent administration of Pseudoephedrine and Furazolidone, they should not be taken together.

Because of its Pseudoephedrine content, Windexco Syrup may partially reverse the hypotensive action of drugs which interfere with sympathetic activity including Bretylium, Bethanidine, Guanethidine, Debrisoquine, Methyl dopa and  $\beta$ -adrenergic blocking agents.

**Undesirable Effects:**

More common side effects are drowsiness or insomnia and gastrointestinal upset. Other less common side effects include dizziness, excitation, sweating, thirst, tachycardia, precordial pain, palpitations, difficulty in micturition, hallucinations and convulsions. A fixed drug eruption due to Pseudoephedrine, taking the form of erythematous nummular patches, has been reported but should be regarded as a rare occurrence.

**Storage:**

Store below 30°C. Protect from light. Keep cap tightly closed.

Keep out of reach of children. Jauhkan daripada capaian kanak-kanak.

**Shelf Life:**

3 years from the date of manufacture.

Manufacturer and Product Registration Holder

**WINWA MEDICAL SDN. BHD.**

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Size: 85mm x 165mm

■ Dark Blue (Pantone 2767U)